

Math 514, F13
Final Exam

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 10 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	12	
2	10	
3	10	
4	12	
5	13	
6	8	
7	8	
8	10	
9	10	
10	12	
Total	105	

Good Luck !

Problem 1. (12 points)

(a) Two dice are rolled fair. Let A be the event that the total of the scores is 8 or higher. Let B be the event that the scores on the two dice are different. Find $P(A)$, $P(B)$, $P(A|B)$, and $P(B|A)$.

$$P(A) = \frac{15}{36} = \frac{5}{12}$$

$$P(A \cap B) = \frac{12}{36} = \frac{1}{3}$$

$$P(B) = 1 - \frac{6}{36} = 1 - \frac{1}{6} = \frac{5}{6}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{12}{36}}{\frac{5}{6}} = \frac{2}{5}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{12}{36}}{\frac{5}{12}} = \frac{4}{5}$$

(b) A and B are two independent events with $P(A) = 0.9$, $P(B) = 0.7$. Find $P(A \cup B)$, $P(B|A)$, and $P(B \setminus A)$.

$$P(A \cap B) = P(A)P(B) = 0.9 \times 0.7 = 0.63.$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.9 + 0.7 - 0.63 = 0.97. \end{aligned}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.63}{0.9} = 0.7$$

$$\begin{aligned} P(B \setminus A) &= P(B) - P(A \cap B) \\ &= 0.7 - 0.63 = 0.07. \end{aligned}$$

Problem 2. (10 points) Let $f(x) = 6x - 6x^2$ for $0 \leq x \leq 1$ and $f(x) = 0$ otherwise. Verify that $f(x)$ is a probability density function and find $E[X]$ for a random variable X whose p.d.f. is $f(x)$.

i). $f(x) \geq 0$,

ii) $\int_{-\infty}^{+\infty} f(x) dx = \int_0^1 6x - 6x^2 dx = [3x^2 - 2x^3]_{x=0}^{x=1} = 1 - 0 = 1$
 $\Rightarrow f$ is a p.d.f.

$$E[X] = \int_{-\infty}^{+\infty} x f(x) dx = \int_0^1 x(6x - 6x^2) dx$$

$$= \int_0^1 6x^2 - 6x^3 dx = [2x^3 - \frac{3}{2}x^4]_{x=0}^{x=1} = 2 - \frac{3}{2} = \frac{1}{2}.$$

Problem 3. (10 points) Suppose the movement of a stock follows: if its present price is s , then after one time period, the price will be either $2s$ with probability $1/2$ or $s/2$ with probability $1/2$. Assume that the successive movements are independent, approximate the probability that the stock price will be at most 50% of present price after the next 1000 time periods.

$$X_i = \frac{S(i)}{S(i-1)} = \begin{cases} 2 & \text{with } P = \frac{1}{2} \\ \frac{1}{2} & \text{with } 1 - P = \frac{1}{2} \end{cases}$$

$$P\left\{\frac{S(1000)}{S(0)} < 0.5\right\} = P\left\{\frac{S(1)}{S(0)} \cdots \frac{S(1000)}{S(999)} < 0.5\right\}$$

$$= P\{X_1 \cdot X_2 \cdots X_{1000} < 0.5\}$$

$$= P\left\{\sum_{i=1}^{1000} \log(X_i) < \log 0.5\right\}$$

$$E[\log(X_i)] = \frac{1}{2} \cdot \log 2 + \frac{1}{2} \log \left(\frac{1}{2}\right) = \frac{1}{2} \log 1 = 0.$$

$$\text{Var}(\log(X_i)) = E[(\log(X_i))^2] - E[\log(X_i)]^2$$

$$= E[(\log(X_i))^2] = \frac{1}{2} \cdot (\log 2)^2 + \frac{1}{2} \cdot (\log \frac{1}{2})^2 = (\log 2)^2$$

$$\Rightarrow \text{By CLT, } \sum_{i=1}^{1000} \log(X_i) \rightarrow N(0, 1000(\log 2)^2)$$

$$\Rightarrow P\left(\frac{S(1000)}{S(0)} \leq 0.5\right) = P\left(Z \leq \frac{\log 0.5}{\sqrt{1000} \log 2}\right) = P(Z \leq -0.0316) \approx 0.4874.$$

Problem 4. (12 points) Let $S(t)$, $t \geq 0$ be a geometric Brownian motion process with drift parameter $\mu = 0.2$ and volatility parameter $\sigma^2 = 0.36$. Find

(a) $P(S(3) < S(1) < S(0))$.

$$\begin{aligned}
 &= P(S(3) < S(1) \text{ \& \& } S(1) < S(0)) \\
 &= P(S(3) < S(1)) \cdot P(S(1) < S(0)) \\
 &= P\left(\frac{S(3)}{S(1)} < 1\right) \cdot P\left(\frac{S(1)}{S(0)} < 1\right) \\
 &= P(X(2) < 0) P(X(1) < 0) \\
 &= P\left(Z < \frac{0 - \mu \cdot 2}{\sigma \sqrt{2}}\right) P\left(Z < \frac{0 - \mu \cdot 1}{\sigma \sqrt{1}}\right) \\
 &= P\left(Z < -\frac{0.4}{0.6\sqrt{2}}\right) P\left(Z < \frac{-0.2}{0.6}\right) \\
 &= 0.3187 \times 0.3694 \\
 &\approx 0.1177.
 \end{aligned}$$

(b) $E[S(2)]$ and $\text{Var}(S(2))$ given $S(0) = 10$.

$$\begin{aligned}
 S(2) &= S(0)e^{X(2)} = 10e^{X(2)} \\
 E[X(2)] &= \mu \cdot 2 = 0.4, \quad \text{Var}(X(2)) = \sigma^2 \cdot 2 = 0.72. \\
 E[S(2)] &= S(0) E[e^{X(2)}] \\
 &= 10 E[e^{X(2)}] = 10 e^{E[X(2)] + \frac{1}{2} \text{Var}(X(2))} \\
 &= 10 \cdot e^{0.4 + 0.72/2} = 10 e^{0.76} \\
 &= 21.3828. \\
 \text{Var}(S(2)) &= E[S(2)^2] - E[S(2)]^2 \\
 &= E[100 e^{2X(2)}] - (10 e^{0.76})^2 \\
 &= 100 e^{0.8 + 1.44} - 100 e^{1.52} \\
 &\approx 482.11.
 \end{aligned}$$

Problem 5. (13 points)

(a) A loan for \$10,000 is repaid over 60 months with the first payment due one month from now. The annual nominal interest rate is 7.2% compounded monthly. Find the monthly payment.

$$\begin{aligned} 10000 &= \frac{A}{1+r} + \dots + \frac{A}{(1+r)^{60}} \\ 10000 &= \frac{A}{1+r} \left(1 + \frac{1}{1+r} + \dots + \frac{1}{(1+r)^{59}} \right) \\ &= \frac{A}{1+r} \frac{1 - \frac{1}{(1+r)^{60}}}{1 - \frac{1}{1+r}} \\ &= A \frac{1 - \frac{1}{(1+r)^{60}}}{r} \\ \Rightarrow A &= 10000 \frac{r}{1 - \frac{1}{(1+r)^{60}}} \approx 198.96 \end{aligned}$$

$r = \frac{0.072}{12} = 0.006$

(b) Income streams $a = (70, 60, 60, 40)$ and $b = (60, 80, 40, 50)$ consist of income received at the end of each of the next four years. Which income stream is preferred (regardless of the interest rate)? State clearly the test that you are using.

$$\begin{aligned} a = (70, 60, 60, 40) &\Rightarrow A = (70, 70+60, 70+60+60, 70+60+60+40) \\ &= (70, 130, 190, 230) \end{aligned}$$

$$\begin{aligned} b = (60, 80, 40, 50) &\Rightarrow B = (60, 60+80, 60+80+40, 60+80+40+50) \\ &= (60, 140, 180, 230) \end{aligned}$$

$$A_1 = 70 > 60 = B_1, \quad A_1 + A_2 = 200 = 200 = B_1 + B_2$$

$$A_1 + A_2 + A_3 = 390 > 380 = B_1 + B_2 + B_3$$

$$A_1 + A_2 + A_3 + A_4 = 620 > 610 = B_1 + B_2 + B_3 + B_4$$

$\Rightarrow a$ is preferred regardless of the interest rate.

Problem 6. (8 points) Consider two European put options on the same security, both of which have the same expiration time t . Suppose the exercise prices of the two puts are K_1 and K_2 where $K_1 \geq K_2$, and $P_1 \geq P_2$ where P_i is the price of the put with strike K_i , $i = 1, 2$. Argue that $K_1 - K_2 \geq P_1 - P_2$.

If $K_1 - K_2 < P_1 - P_2$, then we can always make money by selling P_1 and buying P_2 .

Problem 7. (8 points) The spot rate for the Australian dollar is \$0.7784, the six month forward rate is \$0.77472, and the US interest rate is 5.05% per year. Estimate the Australian interest rate.

$$F = S e^{(r_u - r_a)t}$$

$$0.77472 = 0.7784 e^{(0.0505 - r_a) \cdot \frac{1}{2}}$$

$$\log \frac{F}{S} = (r_u - r_a)t$$

$$r_a = r_u - \frac{1}{t} \log \left(\frac{F}{S} \right)$$

$$= 0.0505 - 2 \times \log \left(\frac{0.77472}{0.7784} \right)$$

$$= 0.06$$

Problem 8. (10 points) An experiment can result in any of the outcomes 1, 2, or 3. If there are ~~three~~ different wagers with

$$\begin{aligned} r_1(1) &= 3, r_1(2) = -8, r_1(3) = 0, \\ r_2(1) &= 5, r_2(2) = x, r_2(3) = -2, \\ r_3(1) &= 10, r_3(2) = 0, r_3(3) = -5. \end{aligned}$$

What must x be equal to if there is no arbitrage?

There must be P_1, P_2, P_3 such that

$$\begin{cases} 3P_1 - 8P_2 = 0 & - \textcircled{1} \\ 5P_1 + xP_2 - 2P_3 = 0 & - \textcircled{2} \\ 10P_1 - 5P_3 = 0 & - \textcircled{3} \\ P_1 + P_2 + P_3 = 1 & - \textcircled{4} \end{cases}$$

$$\textcircled{1} \Rightarrow P_2 = \frac{3}{8}P_1, \quad \textcircled{3} \Rightarrow P_3 = 2P_1$$

thus we get by putting above in $\textcircled{4}$

$$P_1 + \frac{3}{8}P_1 + 2P_1 = 1$$

$$\frac{27}{8}P_1 = 1 \Rightarrow P_1 = \frac{8}{27}$$

$$\Rightarrow P_2 = \frac{3}{8} \cdot \frac{8}{27} = \frac{1}{9}, \quad P_3 = 2 \cdot \frac{8}{27} = \frac{16}{27}$$

Put everything into $\textcircled{2}$, we obtain.

$$5 \cdot \frac{8}{27} + x \cdot \frac{1}{9} - 2 \cdot \frac{16}{27} = 0$$

$$\frac{x}{9} = \frac{32}{27} - \frac{40}{27}$$

$$\frac{x}{9} = -\frac{8}{27}$$

$$\Rightarrow x = -\frac{8}{3}$$

Problem 9. (10 points) On November 3, 2006 the price of Exxon-Mobil stock was \$72.10 per share, its annual volatility parameter was estimated to be $\sigma = 0.18411$, and the annual yield on the three-month treasury bill was $r = 4.93\%$. There were 53 trading days to the expiration of the January, 2007 options contracts. Assuming that there are 252 trading days in the year, calculate the no-arbitrage price for the call option with a strike price of \$75.00 using the Black-Scholes formula.

$$S = 72.10, \quad K = 75, \quad t = \frac{53}{252} \approx 0.2103, \quad r = 0.0493, \quad \sigma = 0.18411$$

$$d = \frac{rt + \sigma^2 t / 2 - \ln(K/S)}{\sigma \sqrt{t}} \approx -0.302$$

$$\begin{aligned} \Rightarrow C &= S \Phi(d) - Ke^{-rt} \Phi(d - \sigma \sqrt{t}) \\ &= 72.10 \Phi(-0.302) - 75e^{-0.0493 \times 0.2103} \Phi(-0.302 - 0.18411 \sqrt{0.2103}) \\ &= 1.5449. \end{aligned}$$

Problem 10. (12 points) The prices of a security follow a geometric Brownian motion with drift parameter $\mu = 0.06$ and volatility parameter $\sigma = 0.4$. The security's present price is 40. A new investment that is being marketed costs C ; the investment will pay 100 at the end of 1 year either if the price of the security at 6 months is at least 42 or if the price of the security at 1 year is at least 5 percent above its price at 6 months, and will pay 0 otherwise. The nominal interest rate is 6%, continuously compounded.

(a) What must be the value of C if this new investment is not to give rise to an arbitrage?

The risk neutral drift parameter $r - \sigma^2/2 = 0.06 - 0.4^2/2 = -0.02$

Let W be the normal random variable with drift $\mu \pm = -0.02 \times \frac{1}{2} = -0.01$
and sd. $\sigma\sqrt{t} = 0.4\sqrt{\frac{1}{2}} \approx 0.2828$.

$$\begin{aligned} C &= e^{-0.06} \cdot 100 \cdot P\left(\left\{\frac{S(1/2)}{S(0)} \geq \frac{42}{40}\right\} \cup \left\{\frac{S(1)}{S(1/2)} \geq 1.05\right\}\right) \\ &= e^{-0.06} \cdot 100 \cdot [P(W > \log(1.05)) + P(W > \log(1.05)) - P^2(W > \log(1.05))] \\ &= e^{-0.06} \cdot 100 [2P(Z > \frac{\log(1.05) + 0.01}{0.2828}) - P^2(Z > \frac{\log(1.05) + 0.01}{0.2828})] \\ &= e^{-0.06} \cdot 100 [2 \times 0.4177 - 0.4177^2] \\ &\approx 62.24. \end{aligned}$$

(b) What is the probability the investment makes money for its buyer?

Let W be the ^{normal} random variable with drift $\mu t = 0.06 \times \frac{1}{2} = 0.03$
and sd. $\sigma\sqrt{t} = 0.2828$.

$$\begin{aligned} P\left(\frac{S(1/2)}{S(0)} \geq 1.05\right) &= P(W > \log(1.05)) \\ &= P(Z > \frac{\log(1.05) - 0.03}{0.2828}) \\ &\approx 0.4735 \end{aligned}$$

$$\Rightarrow P = 2 \times 0.4735 - 0.4735^2 \approx 0.7228$$